

SPEC-DRIVEN INTERVIEWS

Escape the syntax trap. Stop guessing. Start engineering.

Most software professionals prepare for coding interviews by memorizing hundreds of random LeetCode challenges, only to freeze when presented with a modified problem in a high-pressure interview. They fall into the *syntax trap*—writing code immediately, guessing changes to fix failing assertions, and patching edge cases until the code becomes an unmaintainable mess.

The **Spec-Driven Coding Interviews** series introduces a revolutionary, platform-agnostic paradigm that treats technical interviews like rigorous engineering specifications. Instead of typing blindly, you will learn to construct software by defining mathematical invariants, systematically decomposing complex problems, and proving correctness *before* writing a single line of code.

In this book, you will master:

- * **The Invariant-First Strategy:** A structured interview framework to establish execution boundaries and prove correctness under pressure.
- * **Algorithmic Mastery & 24 Canonical Patterns:** Deep dives into sliding windows, prefix sums, DP, graph traversals, and 20 timed mock assessments.
- * **Low-Level Runtime Performance:** Memory architecture (GC, heaps, virtual threads, GIL), CPU cache locality, and zero-allocation optimization.
- * **System Design & Architecture:** Consistent hashing, CAP theorem, CQRS, event streaming, rate limiters, database sharding, and distributed transactions.

Tailored with 100% native, language-specific implementations. Demonstrate industry-grade technical leadership and command your interview.

Harinath Mallepally

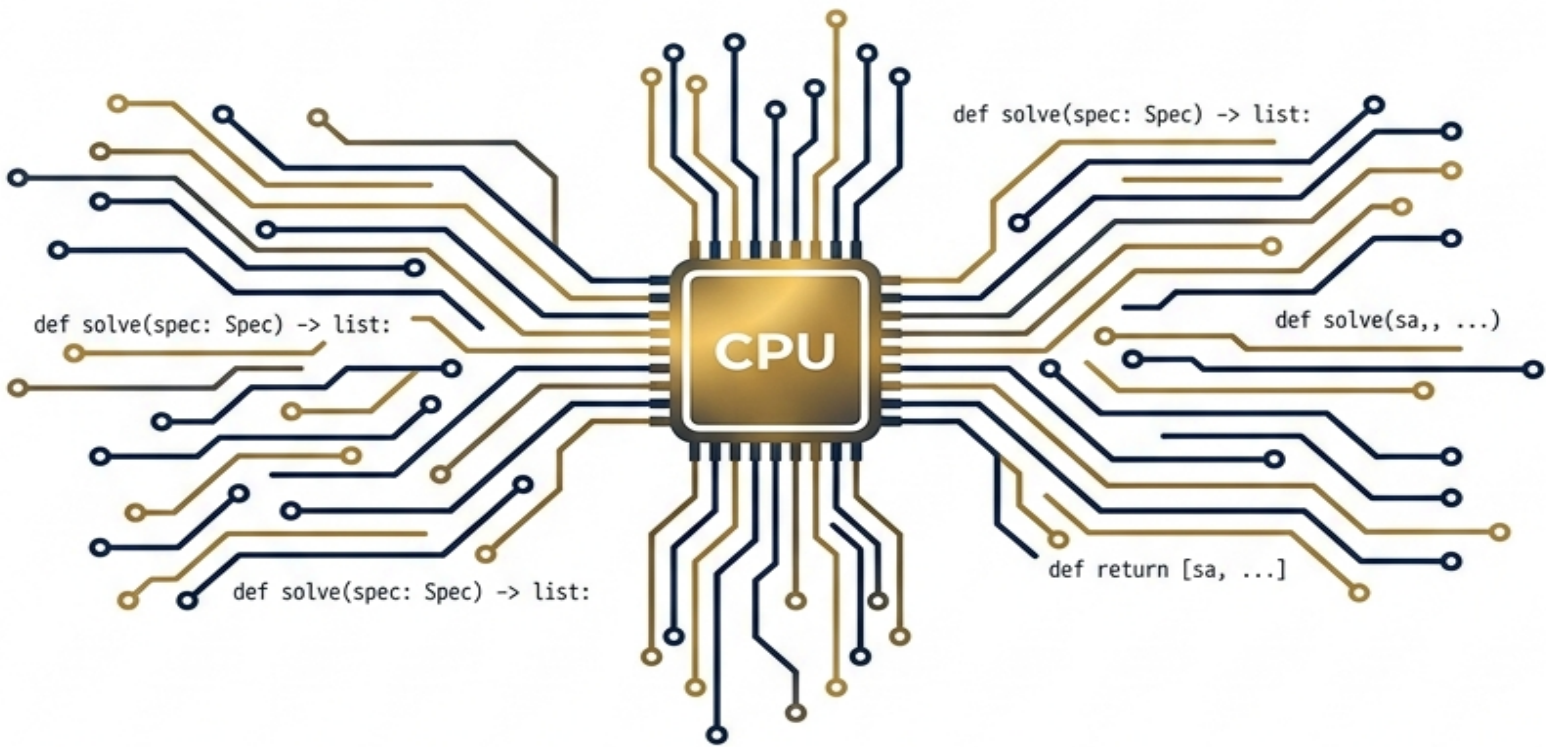
Spec-Driven Coding Interviews - Python Edition

THE DEFINITIVE GUIDE TO MASTERING
SPEC-DRIVEN INTERVIEWS

Spec-Driven Coding Interviews

Python Edition

The Advanced Reference Manual
for Software Professionals



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